

MCO2: Traceroute Tool using ICMP

Test Cases:

[google.com](https://www.google.com)

```
=== ICMP Traceroute Utility ===
[1] google.com
[2] dlsu.instructure.com
[3] www.dlsu.edu.ph
[4] All
[5] Exit
Enter your choice (1-5): 1

Traceroute to google.com:
1 rtt=1 ms 192.168.254.254
* * * Request timed out.
2 rtt=5 ms 192.168.0.1
* * * Request timed out.
* * * Request timed out.
* * * Request timed out.
* * * Request timed out.
5 rtt=8 ms 142.250.174.150 [Mountain View, California, United States] (ISP: Google LLC)
6 rtt=6 ms 142.251.233.215 [Mountain View, California, United States] (ISP: Google LLC)
7 rtt=4 ms 72.14.232.215 [Mountain View, California, United States] (ISP: Google LLC)
8 rtt=6 ms 172.217.27.78
```

dlsu.instructure.com

```
=== ICMP Traceroute Utility ===
[1] google.com
[2] dlsu.instructure.com
[3] www.dlsu.edu.ph
[4] All
[5] Exit
Enter your choice (1-5): 2

Traceroute to dlsu.instructure.com:
1 rtt=1 ms 192.168.254.254
* * * Request timed out.
* * * Request timed out.
* * * Request timed out.
* * * Request timed out.
* * * Request timed out.
* * * Request timed out.
* * * Request timed out.
* * * Request timed out.
* * * Request timed out.
* * * Request timed out.
* * * Request timed out.
* * * Request timed out.
* * * Request timed out.
* * * Request timed out.
* * * Request timed out.
9 rtt=6 ms 13.33.151.26 [New York, New York, United States] (ISP: AWS CloudFront (GLOBAL))
```

www.dlsu.edu.ph

```
=== ICMP Traceroute Utility ===
[1] google.com
[2] dlsu.instructure.com
[3] www.dlsu.edu.ph
[4] All
[5] Exit
Enter your choice (1-5): 3

Traceroute to www.dlsu.edu.ph:
1 rtt=1 ms 192.168.254.254
* * * Request timed out.
* * * Request timed out.
* * * Request timed out.
* * * Request timed out.
* * * Request timed out.
5 rtt=8 ms 120.28.9.254 [Manila, Metro Manila, Philippines] (ISP: Mandarin BIZ)
6 rtt=8 ms 104.20.42.168 [Toronto, Ontario, Canada] (ISP: Cloudflare, Inc.)
```

Declaration of Tools and AI Use

1. **Google Gemini 3.1**
2. **GPT Sol 5.6**
3. **Claude Sonnet 5**

All LLMs were used for analyzing the project specifications, identifying remaining tasks, completing the host execution logic, explaining ICMP protocol mechanics regarding firewalls and dropped packets, and troubleshooting cross-platform (Windows/Linux) raw socket behaviors.

AI Prompts Used

1. *"Can you explain the structure of an ICMP Time Exceeded packet and how to parse it from the IP header?"*
2. *"How do I calculate the ICMP header checksum correctly in Python 3 using the struct module?"*
3. *"Why do raw sockets require Administrator/root privileges?"*
4. *"Running py traceroute.py.. all my requests just time out, why is that? Could it be a firewall issue?"*
5. *"How do I extract the exact round-trip time (RTT) from the struct.unpack payload?"*
6. *"Ran this on my VPS [output showing timeouts and successful final hop] is this normal? Why do some routers drop the ICMP packets?"*
7. *"This is our ICMP traceroute implementation for a networking class assignment.... Don't assume it's correct; review it critically as a network engineer would."*

References

1. Postel, J. (1981). *Internet Control Message Protocol* (RFC 792). IETF. <https://datatracker.ietf.org/doc/html/rfc792>
2. Python Software Foundation. (n.d.). *socket — Low-level networking interface*. Python 3 Documentation. <https://docs.python.org/3/library/socket.html>
3. Python Software Foundation. (n.d.). *struct — Interpret bytes as packed binary data*. Python 3 Documentation. <https://docs.python.org/3/library/struct.html>
4. IP-API.com. (n.d.). *JSON API Documentation*. IP-API. <https://ip-api.com/docs/api:json>
5. Microsoft. (n.d.). *Windows Defender Firewall with Advanced Security*. Microsoft Learn. <https://learn.microsoft.com/en-us/windows/security/operating-system-security/network-security/windows-firewall/>